Lab 7: STRUCTURE AND UNION

Objective
- To be familier with structure and union in C

Program#1

Problem Analysis

To create a program in C to create structure named student Having member roll no. and name. Read a record of one student and display it.

ALGORITHM:

1. Start
2. Define a structure student with members roll no and name.
3. Declare a structure variable.
4. Read structure details (Roll-no. and name)
5. Store the input.
6. Display the student details
7. STOP

C-Program:

```
# include <stdio.h>
 Struct Student
 {
  int rollno.
  Char name [20],
  };
 int main ( )
 {
 Struct student S = {32, Meghan Adhikari}
 Print f ("Student details is : \n);
 Print f ("Roll %d\t, Name: %s\n", S.roll no, S.name);
  return 0;
  }
```

OUTPUT:

Student detail is:

Roll no: 18 Name: Meghan Adhikari

RESULT:

Student detail was printed sucessfully

PROGRAM #2

PROGRAM ANALYSIS

This program uses a structure named student to store details of multiple student. To create a structure named student having member roll-no and name records of n students and display records having roll-no greater than 20.

ALGORITHM:

1. Start
2. Define a structure student with members roll no. and name
3. Read number of students n
4. Declare an array of structure of size n
5. for each student (i=0 to n-1);
6. Read and store in array
7. If roll no > 20 then display detail
8. Stop

C-PROGRAM:

```
# include <stdio.h>
type def struct
 {
  int roll no;
Char name [20];
 } student;
int main( )
 {
Student s [50];
int i,n;
printf ("Enter n:"),
```

```
Scanf("%d",&n);
Printf("Enter details of student:\n");
for(i=0;i<n;i++)
{
printf("Enter roll no:");
Scanf("%d",&s[i].rollno);
Printf("Enter name:");
fflush(stdin);
gets(s[i].name);
}
printf("\n students detail having roll no. greater than
 20:\n");
for(i=0;i<n;i++)
{
if(s[i].rollno>20)
}
Printf("Roll. %d\t Name: %s\n",s[i].roll-no,s[i].name;
}}
return 0;
}
```

RESULT:

Details of n student ~~of~~ was displayed sucessfully.

PROGRAM#3

PROGRAM ANALYSIS:

A program in C to create a structure named book having member name, price and author. Read records of n books and sort records based on price in ~~according~~ ascending order we use the concept of structure & union to create this C-program.

ALGORITHM:

1. Start
2. Define a structure book and members
3. Read n
4. Declare an array or structures of size n
5. Read book and store in array.

6. Short the books in ascenting order of price using a shorting method (eg. bubble short)
   - compare
   - swop
7. Display shorted book records
8. stop.

C-Program :

```
# include <stdio.h>
type def stouct
{
Char name [20];
int price;
Char author [40];
} book;
int main ( )
{
book b[50], temp;
int i,j,n
Print f ("Enter n );
Scanf ("%d", &n);
Print f("Enter detail of books :\n);
for (i=0; i<n; i++)
{print ("Enter book name:);
fflush (stdin);
gets (b[i].name);
print f ("Enter price");
Scan f ("%d, &b [i].name : ");
fflush (stdin);
gets (b [i].author);
}
for (i=0; i<n -1; i++)
{
for (j = i+1; j<n; j++)
{
if [b(i).price >b(j).pre)
{
temp = b[i]
```

```
b[j] = temp;
}}}
printf("shorted records are :\n");
for (i=0; i<n; i++)
{
printf("Book name = %s\t price = %d\t Author: %s\n", b[i].name, b[i].price, b[i].author);
}
return 0;
}
```

OUTPUT:-

Enter n:2
Enter details of books:
Enter book name : Physics
Enter price : 650
Enter author name : Hari
Enter book name : ~~English~~ Math
Enter price : 80
Enter author : Shyam
Shared records are

Bookmame = Math price = 80 Author = Shyam
Bookname = Physics price = 650 Author = Hari.

Result:

Details of book was printed successfully

PROGRAM 4:

PROGRAM ANALYSIS:

To write in C to check add to complex numbers using structure and user defined functions. The result is displayed in proper format. This program demonestrate the use of structures along with functions and passing structured data between functions.

ALGORITH:

1. Start
2. Define structure complex with member real & img
3. Declare two structures
4. Read real & img part
5. Create function for addition
6. Store the result in structure variable.
7. Display the result
8. STOP

C-Program

```
# include <stdio.h>
type of struct.
{
 int. real;
 int. img;
 } complex;
Complex ~~numb~~ resut;
result: real = c1.real + c2+real
result. img = c1.~~rag~~ img + c2. img
return result;
}
int main ( )
{
complex num1, num2 , Sum;
printf ("Enter real & img. part of num 1: ");
Scanf ("%d %d", & num1. real, & num1 :img );
printf ("Enter real & img. part of num 2 : ");
Scanf ("%d %d", & num2. real, & num2. img );
```

```
Sum = add (num 1, num 2);
printf("Sum = %d + %di\n", sum.real, sum.img);
return 0;
}
```

OUTPUT

Enter real & img part of num. 1 : 2 3
Enter real & img part of num 2 : 4 5
Sum = 6+8i

RESULT

Output was displayed successfully

CONCLUSION:

In this lab session we learned to use the structure and it along with user defined functions to operature various programs in C successfully).